University Library Visitor Registration and Daily Counting System

Course: PROG101 Principles of Programming Logic and Design
Assignment: Assignment 1: Designing Logical Solutions for Sierra Leone Community Challenges
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Overview

University libraries often record visitors in paper logbooks. Entries can be illegible or incomplete, and counting the visits for a day has to be done by hand. This project is a **logical design** (no source code) for a simple menu-driven system that registers library visitors and counts the visits for any chosen date.

The design uses an algorithm, pseudocode, a flowchart and modular design, with decision structures, repetition structures, constants, variables, data types, Boolean logic and a function.

The system

The menu has three options:

1. Register Visitor
2. View Daily Records
3. Exit

- **Register Visitor:** captures student name, student ID and visit date. A record with a blank name or ID is rejected ("Name and Student ID are required"); otherwise it is saved ("Visitor Registered Successfully").
- **View Daily Records:** asks for a date, displays every saved record for that date, and shows "Total Visits". If there are none it shows "No Records Available".
- **Exit:** shows "System Closed" and ends the program. Any other entry shows "Invalid Option".

Modules

Module	Role
`Main`	Program control: WHILE loop and IF / ELSE IF routing

Module	Role
`DisplayMenu`	Shows the three options
`RegisterVisitor`	Collects, validates and saves a visit
`IsValidVisitor`	Function that returns a Boolean (name and ID not blank)
`ViewDailyRecords`	Searches the records by date, displays and counts them

Repository contents

Folder	Contents
`algorithm/`	Step-by-step algorithm (`algorithm.md`)
`pseudocode/`	Structured pseudocode with modules and a function (`pseudocode.txt`)
`flowcharts/`	Flowgorithm flowchart image (PNG) and project file (.fprg)
`report/`	The full assignment report (Word document)

Sustainable Development Goals

- **SDG 4: Quality Education** (primary): better management of the library as a learning facility.
- **SDG 16: Peace, Justice and Strong Institutions:** reliable, transparent records for management and funders.
- **SDG 9: Industry, Innovation and Infrastructure:** a low-cost digital tool for a manual process.

Open source and Digital Public Goods

The design is released under the MIT licence (see `LICENSE`) so other institutions can reuse and improve it. The report explains how it could meet the Digital Public Goods Standard.

Tools

- Flowchart built and tested in Flowgorithm.
- Version control with Git and GitHub.

Note

This is a conceptual prototype. `visitorRecords` represents the list of visit records. A real implementation would store records in a file or database so they remain after the program closes.